

Semianalytic Reconstruction of the Mean-Field Gain Model for ^{85}Rb Double- Λ Four-Wave Mixing

Separation of semiclassical gain, quantum fluctuations, and external detection loss

GABES model-audit report

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Revision status. The semiclassical mean-field chain is retained and corrected. The former gain-only squeezing values, -8.10 dB and -15.62 dB, are invalidated as physical predictions because the assumed two-mode map gives $[\hat{a}_s^{\text{out}}, \hat{a}_s^{\text{out}\dagger}] = 0.623$ rather than one. This revision specifies a commutator-preserving quantum-Langevin model and tests a *minimum-vacuum mathematical dilation* of the static drift. The microscopic atomic diffusion matrix $D(\Omega_{\text{SA}})$ is not present in the current code, so no physically validated squeezing spectrum is claimed.

Abstract

The report reconstructs the warm-vapor ^{85}Rb D1 double- Λ four-wave-mixing calculation in three explicitly separated layers: (i) a semiclassical atomic-response and mean-field gain model, (ii) a quantum-fluctuation propagation model with Langevin reservoirs, and (iii) an external detection model. The pump-only state is formulated as a finite-dimensional Liouvillian null-space problem, weak fluctuations are resolved at an independent spectrum-analyzer frequency $\Omega_{\text{SA}}/2\pi \simeq 0.1\text{--}4$ MHz, and the non-collinear two-photon Doppler shift is derived in the required multidimensional velocity average. The physical susceptibility is dimensionless while the reduced response has dimension time; the Voigt prefactor, conjugate energy relation, and all 2π conventions are corrected.

The constant- M 2×2 matrix exponential and the classical gain-gap identity remain valid mean-field results. They are not, by themselves, a Bogoliubov quantum transformation in a dissipative cell. A new Floquet audit finds that the archived $N_F = 1$ result is not converged: going to $N_F = 2$ changes G_s by 16.8%, changes G_c by 13.0%, and reverses the sign of $G_s - G_c$; $N_F = 2$ and $N_F = 3$ agree. The literature operating point $\Delta/2\pi \simeq +0.9$ GHz, $\delta/2\pi \simeq -8$ MHz, and $T = 121^\circ\text{C}$ is kept distinct from the archived reduced-model point $(\Delta/2\pi, \delta/2\pi, T) = (-1.50$ GHz, -280 MHz, $110^\circ\text{C})$. Agreement with the previous code is classified only as an algebraic and numerical consistency check within shared assumptions.

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Part I

Definitions and conventions

1 Atomic references and frequency definitions

The four hyperfine states are

$$g_1 = 5S_{1/2}(F = 2), \quad g_2 = 5S_{1/2}(F = 3), \quad e_2 = 5P_{1/2}(F' = 2), \quad e_3 = 5P_{1/2}(F' = 3).$$

The optical reference is

$$\omega_{13} \equiv (E_{e_3} - E_{g_1})/\hbar, \quad \omega_{\text{HF}} \equiv (E_{g_2} - E_{g_1})/\hbar.$$

Every detuning and linewidth in an equation is an *angular* frequency. Numerical values are always displayed after division by 2π :

$$\frac{\Delta}{2\pi} = -1.50 \text{ GHz}, \quad \frac{\delta}{2\pi} = -280 \text{ MHz}, \quad \frac{\Gamma}{2\pi} = 5.746 \text{ MHz}.$$

The pump and seed conventions are

$$\Delta = \omega_p - \omega_{13}, \tag{1}$$

$$\omega_s = \omega_p - \omega_{\text{HF}} + \delta, \tag{2}$$

$$\Omega_{\text{beat}} = \omega_p - \omega_s = \omega_{\text{HF}} - \delta, \tag{3}$$

$$\omega_c = 2\omega_p - \omega_s. \tag{4}$$

Thus Ω_{beat} is the optical pump-seed beat that organizes the Floquet harmonics. It is not the spectrum-analyzer frequency.

The independent radio-frequency analysis variable is

$$\frac{\Omega_{\text{SA}}}{2\pi} \simeq 0.1\text{--}4 \text{ MHz}.$$

It labels quantum sidebands at $\omega_s \pm \Omega_{\text{SA}}$ and $\omega_c \mp \Omega_{\text{SA}}$. With the Fourier convention used here,

$$\delta \hat{a}_j(z, t) = \int_{-\infty}^{\infty} \frac{d\Omega_{\text{SA}}}{2\pi} \delta \hat{a}_j(z, \Omega_{\text{SA}}) e^{-i\Omega_{\text{SA}} t}. \tag{5}$$

Table 1: Frequency definitions used throughout the report.

Symbol	Definition	Reported as	Role
ω_{13}	$(E_{e_3} - E_{g_1})/\hbar$	$\omega_{13}/2\pi$	atomic optical reference
Δ	$\omega_p - \omega_{13}$	$\Delta/2\pi$	pump one-photon detuning
δ	$\omega_s - \omega_p + \omega_{\text{HF}}$	$\delta/2\pi$	Raman detuning
Ω_{beat}	$\omega_p - \omega_s$	$\Omega_{\text{beat}}/2\pi$	optical pump-seed beat
Ω_{SA}	sideband offset from an optical carrier	$\Omega_{\text{SA}}/2\pi$	RF spectrum-analyzer frequency
Γ	population-decay angular rate	$\Gamma/2\pi$	natural linewidth

2 Field, beam, and susceptibility normalization

Canonical traveling-wave amplitudes are photon-flux normalized:

$$[\delta\hat{a}_j(z, t), \delta\hat{a}_k^\dagger(z, t')] = \delta_{jk}\delta(t - t'), \quad P_j = \hbar\omega_j|\alpha_j|^2. \quad (6)$$

If the atomic code uses Rabi amplitudes $\Omega_j = \mathcal{G}_j\alpha_j$, its propagation matrix must be transformed before being interpreted quantum mechanically:

$$M_{\text{can}} = \mathcal{G}^{-1}M_\Omega\mathcal{G}, \quad \mathcal{G} = \text{diag}(\mathcal{G}_s, \mathcal{G}_c^*). \quad (7)$$

The near equality of the two optical wavelengths does not remove this normalization requirement.

The quoted beam sizes

$$w_p = 530 \mu\text{m}, \quad w_s = 330 \mu\text{m}$$

are $1/e^2$ *intensity radii*, not diameters.

The reduced response is defined schematically by $\bar{\chi}_{jk} = \rho_{eg}^{(j \leftarrow k)}/\Omega_k$. Consequently

$$[\bar{\chi}_{jk}] = \text{s}, \quad \chi_{jk} = -\frac{2Nd^2}{\epsilon_0\hbar}\ell_{\text{norm}}\bar{\chi}_{jk}, \quad [\chi_{jk}] = 1. \quad (8)$$

The factor ℓ_{norm} is dimensionless. Its decomposition is audited in Appendix C; the former $0.74 \times 7/144$ factor is inherited, not an exact first-principles result.

3 Operating-point classification

Table 2: Operating points are classified by provenance, not by numerical appearance in a scan.

Point and class	$\Delta/2\pi$	$\delta/2\pi$	T	Meaning
Archived reduced-model optimum	-1.50 GHz	-280 MHz	110°C	Numerical optimum of the deprecated gain-only squeezing objective. It is not an experimental optimum and its $N_F = 1$ result is unconverged.
Sim et al. literature benchmark [1]	$+0.9 \text{ GHz}$	-8 MHz	121°C	Experimentally measured operating point: 600 mW pump, 8–10 μW seed, 12.5 mm cell, $w_p = 530 \mu\text{m}$, $w_s = 330 \mu\text{m}$, and $\theta = 0.32^\circ$.

The literature experiment reports a probe gain near 15–16, a maximum intensity-difference reduction near -7.8 dB , and a squeezing bandwidth near 3.5 MHz. It also reports a noise feature near 2 MHz arising from laser electronics. Those measurements are not properties of the archived $(\Delta/2\pi, \delta/2\pi) = (-1.50 \text{ GHz}, -280 \text{ MHz})$ model point.

Part II

Pump-only atomic steady state

4 Self-consistent finite-dimensional Liouvillian

For each velocity class and longitudinal position, the pump-only state is defined by

$$\mathcal{L}_{\text{pump}}(z, \mathbf{v})\rho_{\text{ss}}(z, \mathbf{v}) = 0, \quad \text{Tr } \rho_{\text{ss}}(z, \mathbf{v}) = 1. \quad (9)$$

A convenient pump rotating frame gives

$$\frac{H_{\text{pump}}}{\hbar} = \omega_{\text{HF}}|g_2\rangle\langle g_2| - \sum_e \Delta_e(\mathbf{v})|e\rangle\langle e| + \frac{1}{2} \sum_{g,e} (\Omega_{ge}^p|e\rangle\langle g| + \Omega_{ge}^{p*}|g\rangle\langle e|), \quad (10)$$

where $\Delta_e(\mathbf{v}) = \Delta_e - \mathbf{k}_p \cdot \mathbf{v}$. Spontaneous emission, collisional optical dephasing, transit relaxation, and ground-state decoherence enter the same Liouvillian. Vectorizing the 4×4 density matrix gives a 16×16 null-space problem; one row is replaced by the trace constraint. Solving that system would make populations and coherences feed back on one another self-consistently.

This equation defines the required *semianalytic pump-only steady state*: the matrix elements are analytic functions of parameters and the finite null space is to be solved numerically. That code migration has not been completed for this audit. All numerical gain tables below therefore retain the former rate/Sylvester pump approximation and are labeled accordingly. Even after implementation, the solution would be exact only within the four-level, Markov, rotating-wave, and specified-rate truncations.

5 Status of the former rate/Sylvester split

The previous report first solved saturation rate equations for populations and then solved a Sylvester equation for coherences. Because the resulting coherences did not feed back into the populations, that construction is not an exact strong-pump steady state. It may be retained only under all of the following assumptions:

- optical coherences can be adiabatically eliminated;
- excited-state interference and coherent population trapping are small;
- pump-induced level dressing is weak compared with relevant separations;
- population redistribution is slow compared with optical dephasing; and
- fitted rates represent the omitted coherent dynamics.

At the archived point the quoted pump Rabi rate is comparable with or larger than the excited-state hyperfine splitting, so these assumptions are not automatically satisfied. All archived population and coherence numbers are therefore classified as a controlled approximation only if a separate error study is supplied.

6 Phenomenological rates and truncations

The four-level pump model uses a Zeeman-averaged hyperfine basis. The $\gamma_{gg}/(2\pi) = 100$ kHz ground-coherence floor, density-dependent collisional term, and optical self-broadening are phenomenological inputs unless independently measured. Their provenance and sensitivity are listed in Appendix B. No calculation involving these quantities is labeled exact.

Part III

Weak-field linear response

7 Frequency-dependent response matrix

Let $\mathcal{V}_k(\Omega_{\text{SA}})$ be the perturbation generated by the weak seed or conjugate sideband. With Eqs. (5) and $\dot{\rho} = \mathcal{L}_{\text{pump}}\rho + \mathcal{V}_{\rho_{\text{ss}}}$,

$$\delta\rho_k(\Omega_{\text{SA}}) = -(\mathcal{L}_{\text{pump}} + i\Omega_{\text{SA}})_{\text{Tr}=0}^{-1} \mathcal{V}_k(\Omega_{\text{SA}})\rho_{\text{ss}}. \quad (11)$$

The inverse acts on the trace-zero subspace. Using the opposite Fourier sign would replace $+i\Omega_{\text{SA}}$ with $-i\Omega_{\text{SA}}$; mixing those conventions is not allowed.

Polarization readout gives the full response matrix

$$\begin{pmatrix} P_s \\ P_c^* \end{pmatrix} = \epsilon_0 \begin{pmatrix} \chi_{ss}(\Omega_{\text{SA}}) & \chi_{sc}(\Omega_{\text{SA}}) \\ \chi_{cs}^*(\Omega_{\text{SA}}) & \chi_{cc}^*(\Omega_{\text{SA}}) \end{pmatrix} \begin{pmatrix} E_s \\ E_c^* \end{pmatrix}. \quad (12)$$

Thus M_q , the reservoir coupling B , and the diffusion matrix D may all depend on Ω_{SA} .

8 Floquet harmonics and convergence requirement

If the chosen rotating frame leaves a periodic pump term at Ω_{beat} , expand

$$\rho(t) = \sum_{n=-N_F}^{N_F} \rho_n e^{-in\Omega_{\text{beat}}t}.$$

The finite block equations have the form

$$(\mathcal{L}_0 + in\Omega_{\text{beat}}) \rho_n + \mathcal{C}_p \rho_{n-1} + \mathcal{C}_m \rho_{n+1} = 0. \quad (13)$$

Trace one is imposed in the $n = 0$ sector and trace zero in every other sector. The truncation is accepted only after N_F convergence of G_s , G_c , $G_s - G_c$, the wrapped phases of χ_{sc} and χ_{cs} , and the specified optimum δ^* .

The audit in Table 7 uses a generalized continued-fraction solver that reproduces the production $N_F = 1$ calculation exactly and agrees with direct finite-block solves to 5.4×10^{-13} . The displayed gains isolate the archived Floquet truncation and still inherit its old dressed- k phase convention; they are *not* corrected Option-A predictions.

9 Non-collinear Doppler geometry

Choose the pump along z and the seed in the x - z plane:

$$\mathbf{k}_p = k_p \hat{\mathbf{z}}, \quad \mathbf{k}_s = k_s (\sin \theta \hat{\mathbf{x}} + \cos \theta \hat{\mathbf{z}}).$$

Then

$$\Delta_{\text{eff}} = \Delta - k_p v_z, \quad (14)$$

$$\delta_{\text{eff}} = \delta - (k_p - k_s \cos \theta) v_z + k_s \sin \theta v_x. \quad (15)$$

The quantitative average is at least two-dimensional:

$$\langle X \rangle_v = \iint \frac{dv_z dv_x}{2\pi\sigma_v^2} \exp\left[-\frac{v_z^2 + v_x^2}{2\sigma_v^2}\right] X(\Delta_{\text{eff}}, \delta_{\text{eff}}). \quad (16)$$

The unused v_y projection integrates to one. For $k_p \simeq k_s = k$,

$$|\mathbf{k}_p - \mathbf{k}_s| = 2k \sin(\theta/2) \simeq k\theta, \quad \frac{\sigma_{\delta,\theta}}{2\pi} \simeq \frac{\theta\sigma_v}{\lambda} = 1.361 \text{ MHz} \quad (17)$$

for $\theta = 0.32^\circ$, $\sigma_v = 193.7 \text{ m/s}$, and $\lambda = 795 \text{ nm}$. These requested values correspond to the archived 110°C estimate. At the 121°C literature point, $\sigma_v \simeq 196.5 \text{ m/s}$ and the width is about 1.38 MHz . Both are much larger than the assumed $\gamma_{gg}/2\pi = 0.10 \text{ MHz}$. The angular term vanishes as $\theta \rightarrow 0$; a roughly 2 kHz collinear residual remains because $\omega_p \neq \omega_s$.

10 Faddeeva base integral and corrected Voigt identity

For a single velocity projection,

$$\left\langle \frac{1}{x_0 - kv + i\Gamma_o} \right\rangle_v = -\frac{i}{k\sigma_v} \sqrt{\frac{\pi}{2}} w\left(\frac{x_0 + i\Gamma_o}{\sqrt{2}k\sigma_v}\right), \quad w(z) = e^{-z^2} \operatorname{erfc}(-iz). \quad (18)$$

With

$$V(x; \sigma, \gamma) = \frac{\operatorname{Re} w[(x + i\gamma)/(\sqrt{2}\sigma)]}{\sqrt{2\pi}\sigma},$$

the dimensionally correct identity is

$$\boxed{\operatorname{Im} \left\langle \frac{1}{x_0 - kv + i\Gamma_o} \right\rangle_v = -\pi V(x_0; k\sigma_v, \Gamma_o).} \quad (19)$$

There is no additional factor $1/k$. Equation (18) is an exact base simple-pole integral; velocity-dependent populations, Raman self-energies, and the two-detuning geometry generally require multidimensional quadrature or a justified partial-fraction reduction. The former one-dimensional Faddeeva check is not a complete treatment of the experimental geometry.

Part IV

Semiclassical field propagation and gain

11 Classical coupled-mode equation

Use the carrier convention $E_j^{(+)} \propto \alpha_j(z) e^{i(\omega_j t - k_{j,z} z)}$. Then the four-wave polarization $E_p^{(+)} E_c^{(+)*}$ has the signal-envelope phase $e^{-i(2k_{p,z} - k_{s,z} - k_{c,z})z}$, and the unrotated envelopes obey

$$\alpha'_s = a\alpha_s + b\alpha_c^* e^{-i\Delta k_{\text{geom}} z}, \quad (\alpha_c^*)' = c\alpha_s e^{+i\Delta k_{\text{geom}} z} + d_0\alpha_c^*,$$

which fixes the phase convention. Define the symmetric rotating frame

$$\tilde{\alpha}_s = \alpha_s e^{+i\Delta k_{\text{geom}} z/2}, \quad \tilde{\alpha}_c^* = \alpha_c^* e^{-i\Delta k_{\text{geom}} z/2}.$$

Direct differentiation then gives

$$\frac{d}{dz} \begin{pmatrix} \tilde{\alpha}_s \\ \tilde{\alpha}_c^* \end{pmatrix} = M_{\text{cl}} \begin{pmatrix} \tilde{\alpha}_s \\ \tilde{\alpha}_c^* \end{pmatrix}, \quad M_{\text{cl}} = Q^{-1} M_E Q, \quad (20)$$

where the Maxwell electric-field matrix is

$$M_E = \begin{pmatrix} \frac{ik_s}{2} \chi_{ss} + \frac{i\Delta k_{\text{geom}}}{2} & \frac{ik_s}{2} \chi_{sc} \\ -\frac{ik_c}{2} \chi_{cs}^* & -\frac{ik_c}{2} \chi_{cc}^* - \frac{i\Delta k_{\text{geom}}}{2} \end{pmatrix}, \quad Q = \operatorname{diag} \left(\sqrt{\frac{2\hbar\omega_s}{\epsilon_0 c A_s}}, \sqrt{\frac{2\hbar\omega_c}{\epsilon_0 c A_c}} \right). \quad (21)$$

Here

$$\Delta k_{\text{geom}} = 2k_{p,z} - k_{s,z} - k_{c,z} \quad (22)$$

uses vacuum carrier wave numbers and geometry only. The effective mode areas A_s, A_c must match the collected modes. Equation (20) is the canonical photon-flux rescaling of the electric-field equation; a Rabi-amplitude implementation must equivalently apply Eq. (7).

No double counting of dispersion

Equation (20) is Option A. The real parts of χ_{ss} and χ_{cc} generate dispersive phase exactly once through the diagonal propagation terms. Refractive indices $n_j = 1 + \operatorname{Re} \chi_{jj}/2$ are *not* inserted again into Δk_{geom} . The archived dressed- k calculation that used both mechanisms is retained only for implementation-history and convergence diagnostics.

12 Constant-matrix exponential and mean-field gains

For an undepleted pump and z -independent coefficients,

$$\begin{aligned} T_{\text{cl}}(L) &= e^{M_{\text{cl}}L} \\ &= e^{sL} \left[\cosh(qL)I + \frac{\sinh(qL)}{q}(M_{\text{cl}} - sI) \right], \end{aligned} \quad (23)$$

$$s = \frac{a+d}{2}, \quad q = \sqrt{\frac{(a-d)^2}{4} + bc}, \quad (24)$$

where a, b, c, d are the entries of M_{cl} . The formula is an algebraic identity for constant M_{cl} . Its physical assumptions are:

- undepleted pump;
- fixed density and temperature;
- no longitudinal pump attenuation;
- z -independent overlap and response coefficients; and
- a single collected transverse mode.

If those assumptions fail, use an ordered product or path-ordered exponential.

For seed-only input,

$$G_s = |T_{\text{cl},11}|^2, \quad G_c = |T_{\text{cl},21}|^2. \quad (25)$$

These are explicitly **semiclassical mean-field gains**. They do not determine a quantum covariance.

13 Classical gain-gap identity

For $u' = au + bv$ and $v' = cu + dv$,

$$\frac{d}{dz}(|u|^2 - |v|^2) = 2\text{Re}(a)|u|^2 - 2\text{Re}(d)|v|^2 + 2\text{Re}[(b - c^*)u^*v]. \quad (26)$$

In a closed lossless parametric model, $\text{Re } a = \text{Re } d = 0$ and $b = c^*$, so $G_s - G_c = 1$. Equation (26) is retained as a property of the classical transfer matrix. If $G_s - G_c \neq 1$, it is evidence that a two-operator quantum map needs additional channels; it is not permission to use a noncanonical Bogoliubov map.

Part V

Quantum fluctuation propagation

14 Why the former two-mode formula fails

The former relation

$$\hat{a}_s^{\text{out}} = \mu \hat{a}_s^{\text{in}} + \nu \hat{a}_c^{\text{in}\dagger}, \quad |\mu|^2 = G_s, \quad |\nu|^2 = G_c$$

implies

$$[\hat{a}_s^{\text{out}}, \hat{a}_s^{\text{out}\dagger}] = G_s - G_c.$$

At the archived representative result this equals 0.623, not one. Therefore the map is invalid without additional noise operators. The formula

$$\text{cov}_0 = \frac{1}{2}[(G_s + G_c) - (G_s - G_c)^2]$$

and every squeezing number derived only from G_s and G_c are removed as general results for the dissipative cell.

15 Quantum-Langevin propagation

Define the Nambu sideband vector

$$\hat{\mathbf{A}}(z, \Omega) = \begin{pmatrix} \delta \hat{a}_s(z, \Omega) \\ \delta \hat{a}_c^\dagger(z, -\Omega) \end{pmatrix}, \quad [\hat{\mathbf{A}}(z, \Omega), \hat{\mathbf{A}}^\dagger(z', \Omega')] = 2\pi J \delta(\Omega - \Omega'), \quad J = \text{diag}(1, -1). \quad (27)$$

The distributed fluctuation equation is

$$\frac{\partial}{\partial z} \hat{\mathbf{A}}(z, \Omega_{\text{SA}}) = M_q(z, \Omega_{\text{SA}}) \hat{\mathbf{A}}(z, \Omega_{\text{SA}}) + B(z, \Omega_{\text{SA}}) \hat{\mathbf{F}}(z, \Omega_{\text{SA}}). \quad (28)$$

Its formal solution is

$$T(z_2, z_1; \Omega) = \mathcal{P} \exp \left[\int_{z_1}^{z_2} M_q(z, \Omega) dz \right], \quad (29)$$

$$\hat{\mathbf{A}}_{\text{out}}(\Omega) = T(L, 0; \Omega) \hat{\mathbf{A}}_{\text{in}}(\Omega) + \int_0^L T(L, z; \Omega) B(z, \Omega) \hat{\mathbf{F}}(z, \Omega) dz. \quad (30)$$

For local Markov reservoirs,

$$[\hat{\mathbf{F}}(z, \Omega), \hat{\mathbf{F}}^\dagger(z', \Omega')] = 2\pi J_F \delta(z - z') \delta(\Omega - \Omega').$$

Canonical preservation requires

$$\boxed{T J T^\dagger + \int_0^L T(L, z) B J_F B^\dagger T^\dagger(L, z) dz = J.} \quad (31)$$

A sufficient local condition is

$$\boxed{M_q J + J M_q^\dagger + B J_F B^\dagger = 0.} \quad (32)$$

16 Covariance and atomic diffusion

Define reservoir symmetrized correlations by

$$\frac{1}{2} \left\langle \{ \hat{\mathbf{F}}(z, \Omega), \hat{\mathbf{F}}^\dagger(z', \Omega') \} \right\rangle = 2\pi N_{\text{res}}(z, \Omega) \delta(z - z') \delta(\Omega - \Omega'), \quad D = B N_{\text{res}} B^\dagger.$$

Then

$$\boxed{V_{\text{out}}(\Omega) = T(\Omega) V_{\text{in}}(\Omega) T^\dagger(\Omega) + \int_0^L T(L, z; \Omega) D(z, \Omega) T^\dagger(L, z; \Omega) dz.} \quad (33)$$

Equation (33) is one 2×2 Nambu block. Before applying the detector model, solve its companion block and assemble

$$\hat{\Xi}(\Omega) = (\hat{a}_s(\Omega), \hat{a}_c(\Omega), \hat{a}_s^\dagger(-\Omega), \hat{a}_c^\dagger(-\Omega))^T$$

with spectral covariance $V_{\Xi}(\Omega)$. The quadrature-sideband covariance used below is

$$V_R(\Omega) = C V_{\Xi}(\Omega) C^\dagger, \quad C = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 1 & 0 \\ -i & 0 & i & 0 \\ 0 & 1 & 0 & 1 \\ 0 & -i & 0 & i \end{pmatrix}, \quad (34)$$

corresponding to $\hat{\mathbf{R}} = (X_s, P_s, X_c, P_c)^T$. This explicit assembly prevents the 2×2 Nambu covariance from being mixed dimensionally with the 4×4 detector matrices. The commutator condition fixes antisymmetric reservoir correlations, not the symmetrized diffusion that determines squeezing. A physical $D(z, \Omega_{\text{SA}})$ must be derived from the pumped atomic Lindblad model, generalized Einstein relations, or an independently justified reservoir model. Independent velocity classes add as independent diffusion powers; one must not square a velocity-averaged noise amplitude.

16.1 Minimum-vacuum mathematical completion

For testing only, any constant drift may be given a quantum-limited local completion. Let

$$K = -(M_q J + J M_q^\dagger) = U \Lambda U^\dagger, \quad B_{\min} = U |\Lambda|^{1/2}, \quad J_F = \text{diag}(\text{sgn } \Lambda), \quad D_{\min} = \frac{1}{2} B_{\min} B_{\min}^\dagger. \quad (35)$$

This construction preserves the commutator and assumes vacuum reservoirs with the minimum compatible symmetrized noise. It is not a derivation of atomic spontaneous-emission noise, and additional diffusion can only be assessed from a microscopic model or data.

At the literature point, an $N_F = 3$, cutoff-converged, Option-A static drift gives, under the diagnostic approximation $Q \propto I$,

$$G_s = 5.6143, \quad G_c = 4.6682, \quad G_s - G_c = 0.9462.$$

The uncompleted drift has a maximum commutator defect 5.51×10^{-2} . The factorized minimum completion reduces the local residual to $4.44 \times 10^{-16} \text{ m}^{-1}$ and the end-to-end relative Frobenius residual to 1.04×10^{-15} . This is a test of Eq. (32), not evidence that $D_{\min}$ equals the atomic diffusion matrix.

17 Frequency-dependent observable and present limitation

The final in-cell observable must be a spectrum $S_-^{\text{out}}(\Omega_{\text{SA}})$, not a scalar. The current microscopic $M_q(\Omega_{\text{SA}})$ and $D(\Omega_{\text{SA}})$ are not implemented. The static minimum completion above is frequency independent and yields only a diagnostic flat curve. It cannot reproduce:

- the squeezing bandwidth;
- low-frequency excess noise;
- the approximately 2 MHz laser/electronic feature;
- frequency-dependent atomic Langevin noise;
- slow-light and group-delay effects; or
- the detector-frequency response.

Part VI

External detection model

18 External loss only after the cell

In the quadrature basis $\hat{\mathbf{R}} = (X_s, P_s, X_c, P_c)^T$, define

$$K_\eta = \text{diag}(\sqrt{\eta_s}, \sqrt{\eta_s}, \sqrt{\eta_c}, \sqrt{\eta_c}), \quad K_\ell = \text{diag}(\sqrt{1 - \eta_s}, \sqrt{1 - \eta_s}, \sqrt{1 - \eta_c}, \sqrt{1 - \eta_c}).$$

Only loss physically located after the cell is applied:

$$V_{R,\text{det}} = K_\eta V_{R,\text{out}} K_\eta^T + K_\ell V_{\text{vac}} K_\ell^T. \quad (36)$$

For symmetric efficiency,

$$V_{R,\text{det}} = \eta_{\text{ext}} V_{R,\text{out}} + (1 - \eta_{\text{ext}}) V_{\text{vac}}.$$

Atomic absorption and spontaneous emission remain distributed inside Eqs. (28)–(33). Cell optical-depth factors $\exp(-\text{OD}_{p,c,\text{seg}})$ are not applied a second time after propagation.

19 Electronic weighting and shot-noise normalization

The linearized photocurrent is

$$\delta\hat{i}_j(\Omega) \propto H_j(\Omega) \left[\alpha_j^* \delta\hat{a}_j(\Omega) + \alpha_j \delta\hat{a}_j^\dagger(-\Omega) \right], \quad \delta\hat{i}_- = \delta\hat{i}_s - w(\Omega) \delta\hat{i}_c. \quad (37)$$

The normalized spectrum is

$$S_-(\Omega) = \frac{\mathbf{m}^\dagger(\Omega) V_{R,\text{det}}(\Omega) \mathbf{m}(\Omega) + S_{\text{el}}(\Omega)}{S_{\text{SNL}}^{(\text{exp})}(\Omega)}. \quad (38)$$

Here $\mathbf{m}(\Omega)$ is the four-component measurement vector obtained from Eq. (37); it contains the detected carrier amplitudes, H_s, H_c , their phases, and the factor $-w$ on the conjugate quadratures. With $V_{\text{vac}} = I/2$ and independent detector vacua, the corresponding ideal electronic shot-noise term is proportional to

$$S_{\text{SNL}}(\Omega) \propto |H_s|^2 \eta_s |\alpha_s|^2 + w^2 |H_c|^2 \eta_c |\alpha_c|^2,$$

before the experiment-specific electronic calibration. The shot-noise reference must use the same detected powers, electronic transfer functions, weight w , resolution bandwidth, and video bandwidth. The report must state whether electronic dark noise is included or subtracted.

Table 3: Loss and noise ledger.

Mechanism	Physical location	Correct treatment
Atomic absorption and spontaneous emission	distributed in the vapor	Complex response in M_q plus associated B and D
Cell windows, filters, and free propagation	after the cell	External beam-splitter efficiency η_{path}
Detector quantum efficiency	detector face	External beam-splitter efficiency η_{QE}
Electronic imbalance and bandwidth	after photodetection	$H_s(\Omega)$, $H_c(\Omega)$, $w(\Omega)$, and calibrated $S_{\text{el}}(\Omega)$
Pump scattering into the detected mode	apparatus dependent	Measured technical-noise spectrum; not a universal atomic-loss formula

The former ansatz

$$\mathcal{N}_{\text{ps}} = \kappa(1 - e^{-\text{OD}_{\text{pump}}})$$

is phenomenological. If retained, it must be written $\mathcal{N}_{\text{ps}}(\Omega) = \kappa(1 - e^{-\text{OD}_{\text{pump}}}) F_{\text{ps}}(\Omega)$ and calibrated to the experimental spectral normalization. It is not used as a universal Langevin noise law in this report.

Part VII

Validation

20 Validation hierarchy

Agreement between the revised algebra and the archived program is an independent reimplementa-
tion of the same reduced model. It is an algebraic and numerical consistency check, not independent
physical validation. The levels below are therefore reported separately.

Table 4: Validation hierarchy. No combined error column is used.

Level	Test performed	Status
Algebraic consistency	Closed-form matrix exponential versus direct exponential; dimensional and frequency audits	PASS
Numerical implementation consistency	Independent Floquet assembly versus archived reduced equations	PASS within shared assumptions
Truncation convergence	$N_F = 1, 2, 3$ and velocity-grid/cutoff studies	$N_F = 1$ FAIL; $N_F = 2$ converged for the tested point
Independent microscopic calculation	Multilevel Zeeman-resolved Liouvillian and atomic diffusion matrix	not yet available
Experimental gain comparison	Reduced model evaluated at the documented ⁸⁵ Rb operating point	performed; substantial discrepancy
Experimental squeezing-spectrum comparison	Full $S_-(\Omega_{SA})$ using microscopic $D(\Omega_{SA})$	not yet available

21 Dimensional-consistency audit

Table 5: Dimensional audit in SI units.

Quantity	Dimension	Check
$\Delta, \delta, \Gamma, \Omega_{SA}, D_e$	s^{-1}	All equations use angular frequency
$\bar{\chi}_{ij}$	s	Contains resolvent factors such as D_e^{-1} ; it is not dimensionless
$Nd^2/(\epsilon_0\hbar)$	s^{-1}	Multiplying $\bar{\chi}$ gives dimensionless χ_{ij}
χ_{ij}	1	Required by $k_j\chi_{ij}/2$ in propagation
M_{cl}, M_q	m^{-1}	Compatible with $\partial_z \mathbf{a} = M\mathbf{a}$
$B\hat{\mathbf{f}}$	field amplitude per metre	Uses delta-correlated reservoir fields in z
$V(x; \sigma, \gamma)$	s	$-\pi V$ has the dimension of an inverse angular frequency
G_s, G_c, S_-	1	Power gains and normalized noise spectra

22 Frequency-convention audit

Table 6: Frequency variables and references.

Symbol	Definition	Convention and role
$\omega_p, \omega_s, \omega_c$	absolute optical angular frequencies	$\omega_c = 2\omega_p - \omega_s$
Δ	$\omega_p - \omega_{13}$	Reported only as $\Delta/(2\pi)$ in GHz or MHz
δ	$\omega_s - \omega_p + \omega_{\text{HF}}$	Reported only as $\delta/(2\pi)$; sign fixes the seed convention
Ω_{beat}	$\omega_p - \omega_s = \omega_{\text{HF}} - \delta$	Optical pump-seed beat; not the analyzer frequency
Ω_{SA}	Fourier frequency of fluctuations	$\Omega_{\text{SA}}/(2\pi) \simeq 0.1\text{--}4\text{ MHz}$
Γ and γ_{gg}	angular decay rates	For example $\Gamma/(2\pi) = 5.75\text{ MHz}$ and $\gamma_{gg}/(2\pi) = 0.10\text{ MHz}$

23 Floquet-truncation convergence

Table 7 repeats the archived-model calculation at the legacy numerical optimum $\Delta/(2\pi) = -1.50\text{ GHz}$, $T = 110^\circ\text{C}$ and approximately $\delta/(2\pi) = -280\text{ MHz}$. It deliberately retains the archived dressed-wave-vector phase convention only to isolate Floquet convergence; because that convention double counts dispersion, these numbers are not the corrected physical prediction.

Table 7: Floquet convergence of the archived reduced model at the fixed point $\delta/(2\pi) = -280\text{ MHz}$. Sectors $n = -N_F, \dots, N_F$ are retained.

N_F	G_s	G_c	$G_s - G_c$	$\arg \chi_{sc}$	$\arg \chi_{cs}$
1	22.5913	21.9687	+0.622609	178.142°	-179.333°
2	19.3415	19.4472	-0.105684	178.173°	-179.366°
3	19.3415	19.4472	-0.105684	178.173°	-179.366°

Table 8: Separate minimization of the archived gain-only objective. All observables in a row are evaluated at that row's δ^* . The scan spacing is 5 MHz.

N_F	$\delta^*/(2\pi)$ [MHz]	G_s	G_c	$G_s - G_c$	$\arg \chi_{sc}$	$\arg \chi_{cs}$
1	-280	22.5913	21.9687	0.622609	178.142°	-179.333°
2	-275	8.97871	8.46840	0.510314	178.402°	-179.582°
3	-275	8.97871	8.46840	0.510314	178.402°	-179.582°

The relative changes from $N_F = 1$ to 2 are

$$\frac{|G_s^{(2)} - G_s^{(1)}|}{G_s^{(2)}} = 16.80\%, \quad \frac{|G_c^{(2)} - G_c^{(1)}|}{G_c^{(2)}} = 12.97\%.$$

The gain gap even changes sign. Thus the original $N_F = 1$ result is *qualitative and unconverged*. The $N_F = 2$ to 3 changes are below $4 \times 10^{-7}\%$ at this tested point. Direct block solves and the independent assembly agree to 5.4×10^{-13} or better. Convergence at one operating point does not prove global convergence; every reported scan must repeat this test.

For the propagation-only Option-A correction at the fixed literature point, the inherited approximate atomic response gives the results in Table 9. Here $N_F = 1 \rightarrow 2$ changes G_s by 0.0863% and G_c by 0.1481%; $N_F = 2$ and 3 agree at the displayed precision. This establishes Floquet convergence of that fixed propagation benchmark, while leaving the pump-state and multidimensional-velocity approximations untouched.

Table 9: Floquet convergence at the fixed literature point $\Delta/(2\pi) = +0.90\text{ GHz}$, $\delta/(2\pi) = -8\text{ MHz}$, and $T = 121\text{ }^\circ\text{C}$. This is not an optimization and does not use the new self-consistent pump state.

N_F	G_s	G_c	$G_s - G_c$	$\arg \chi_{sc}$	$\arg \chi_{cs}$
1	5.609495	4.661245	0.948250	-179.880518°	179.797297°
2	5.614341	4.668160	0.946182	-179.879968°	179.796859°
3	5.614341	4.668160	0.946182	-179.879968°	179.796859°

The inherited weak-response implementation also retains a finite reference seed. Its numerical linearity check is

Reference seed	G_s	G_c	$G_s - G_c$
$2\text{ }\mu\text{W}$	5.614368	4.668743	0.945625
$4\text{ }\mu\text{W}$	5.614359	4.668549	0.945811
$8\text{ }\mu\text{W}$	5.614341	4.668160	0.946182

Between 2 and $8\text{ }\mu\text{W}$, G_s changes by -0.000482% , G_c by -0.012493% , and the gap by $+0.058822\%$. This supports numerical weak-reference linearity at this fixed point only; it does not validate the underlying pump-state approximation.

24 Velocity-integration convergence

The following tests use $N_F = 3$ and the archived one-dimensional Doppler integral. They validate only that integral, not the non-collinear geometry. At a fixed $5\sigma_v$ cutoff, velocity-step convergence is excellent:

Table 10: One-dimensional velocity-step convergence at fixed $|v_z| \leq 5\sigma_v$.

Δv [m/s]	Classes	G_s	G_c	Relative G_s error
10.00	195	18.936590	19.024328	$1.56 \times 10^{-3}\%$
5.00	389	18.936295	19.024021	$2.66 \times 10^{-7}\%$
2.50	777	18.936295	19.024021	$9.58 \times 10^{-8}\%$
1.25	1551	18.936295	19.024021	reference

The cutoff, however, is important:

Table 11: One-dimensional cutoff convergence at $\Delta v = 2.5$ m/s.

Cutoff	G_s	G_c	$G_s - G_c$	Relative G_s error
$3.0\sigma_v$	19.36195	19.46857	-0.10662	2.248%
$3.5\sigma_v$	19.11527	19.21041	-0.09514	0.945%
$4.0\sigma_v$	18.98214	19.07172	-0.08957	0.242%
$4.5\sigma_v$	18.93642	19.02414	-0.08773	$6.4 \times 10^{-4}\%$
$5.0\sigma_v$	18.93629	19.02402	-0.08773	reference

For the experimental crossing angle, a converged quantitative calculation must additionally integrate the mismatch projection in Eq. (16). The angular rms width $\sigma_{\delta,\theta}/(2\pi) = 1.36$ MHz is not negligible relative to $\gamma_{gg}/(2\pi) = 0.10$ MHz.

25 Commutator-preservation test

At the corrected literature operating point, assume for this diagnostic that the collected signal and conjugate have equal effective mode areas and use their near-equal optical frequencies, so $Q \propto I$ in Eq. (21). The resulting static drift in the symmetric mismatch gauge is

$$M_q = \begin{pmatrix} -1.330328 + 242.349727i & -0.410000 + 195.707754i \\ 0.693784 - 195.680004i & 0.341352 - 171.013770i \end{pmatrix} \text{ m}^{-1}.$$

This is the same symmetric gauge as Eq. (21). If the actual collected mode areas or Rabi-to-field factors differ, $Q^{-1}M_EQ$ must be recomputed before applying the canonical metric J . The residuals below are therefore a conditional channel diagnostic, not a completed production normalization audit. For the numerical test only, factor the Hermitian matrix $K = -(M_qJ + JM_q^\dagger)$ as $K = BJ_FB^\dagger$. Here its eigenvalues are 0.189391 m^{-1} and 3.153969 m^{-1} , so a minimum-vacuum dilation with $J_F = I$ exists. This construction proves that the drift can be embedded in a canonical channel; it is not a microscopic derivation of atomic noise.

Table 12: Canonical-commutator audit.

Test	Residual	Result
Old gain-only two-mode map	$ G_s - G_c - 1 = 0.377$	FAIL
Corrected drift with reservoir omitted	$\max TJT^\dagger - J = 5.51 \times 10^{-2}$	FAIL
Local identity with minimum dilation, $Q \propto I$	$\max M_qJ + JM_q^\dagger + BJ_FB^\dagger = 4.44 \times 10^{-16} \text{ m}^{-1}$	PASS conditional
Integrated identity through the cell, $Q \propto I$	$\ TJT^\dagger + \int T_z BJ_FB^\dagger T_z^\dagger dz - J\ _F / \ J\ _F = 1.04 \times 10^{-15}$	PASS conditional
Microscopic $B(\Omega)$ and $D(\Omega)$ over 0.1–4 MHz	Not implemented in the current four-level code	not yet available

Canonical commutator preservation constrains the antisymmetric reservoir correlations. It does not determine the symmetrized diffusion matrix $D(\Omega)$, its temperature, or excess spontaneous-emission noise. Consequently the minimum-vacuum values below are an illustrative diagnostic for one chosen dilation, not a bound and not a Langevin-corrected physical prediction.

26 Corrected model point and experimental comparison

This benchmark is a *propagation-only correction*: it applies Option A and $N_F = 3$ to the inherited rate/Sylvester atomic response. It is not an output of the new pump-null-space and frequency-resolvent formulation in Parts II–III. It uses a $5\sigma_v$ one-dimensional velocity range and the documented point

$$\frac{\Delta}{2\pi} = +0.90 \text{ GHz}, \quad \frac{\delta}{2\pi} = -8 \text{ MHz}, \quad T = 121 \text{ }^\circ\text{C}, \quad L = 12.5 \text{ mm},$$

with 600 mW pump power, $8 \mu\text{W}$ seed power, $w_p = 530 \mu\text{m}$, $w_s = 330 \mu\text{m}$, and $\theta = 0.32^\circ$. The radii are $1/e^2$ intensity radii. The calculation still omits the transverse two-photon Doppler average and therefore remains a reduced-model benchmark. In the same symmetric mismatch gauge used for the commutator audit,

$$T_{\text{cl}} = \begin{pmatrix} -0.376876 + 2.339296i & -0.929014 + 1.950994i \\ 0.931714 - 1.949376i & 1.593377 - 1.777683i \end{pmatrix},$$

which gives $G_s = 5.61434$, $G_c = 4.66816$, and $G_s - G_c = 0.946182$.

Table 13: Separation of mean-field gain and quantum-noise layers at the literature operating point, conditional on the diagnostic $Q \propto I$ normalization. The symmetric external efficiency used in the diagnostic columns is $\eta_{\text{ext}} = 0.8694$.

Layer	G_s	G_c	S_-^{out}	S_-^{det}	Interpretation
Semiclassical reduced model	5.614	4.668	—	—	Mean-field gain only
Ideal Bogoliubov amplifier matched to G_s	5.614	4.614	0.09776 (−10.10 dB)	0.21560 (−6.66 dB)	Lossless limiting case, not the dissipative medium
Minimum-vacuum static dilation	5.614	4.668	0.09230 (−10.35 dB)	0.21084 (−6.76 dB)	Canonical diagnostic with $w = 1$; D is not microscopic
Minimum-vacuum static dilation, DC balanced	5.614	4.668	0.09774 (−10.10 dB)	0.21558 (−6.67 dB)	Same diagnostic with $w = G_s/G_c = 1.20269$
Microscopic quantum-Langevin model	5.614	4.668	not yet available	not yet available	Required physical calculation
Experiment	~ 15.5	apparent 13.6	—	0.166 (−7.8 dB)	Measured spectrum; conjugate ratio is not a separately calibrated transfer coefficient

Table 14: Spectrum-level comparison for the $w = 1$ diagnostic. A constant entry is shown only to expose what the static model cannot reproduce; DC balancing would instead give a flat −6.67 dB.

$\Omega_{\text{SA}}/(2\pi)$	Static minimum-vacuum diagnostic	Microscopic Langevin prediction	Experimental information
0.10 MHz	−6.76 dB	not yet available	Low-frequency technical excess noise is present −8.7 dB ratio of fitted IDS/SQL power slopes at 150 kHz
0.15 MHz	−6.76 dB	not yet available	
2.00 MHz	−6.76 dB	not yet available	Approximately 2 MHz laser-electronics feature
4.00 MHz	−6.76 dB	not yet available	Beyond the approximately 3.5 MHz squeezing bandwidth

The static gain model cannot reproduce squeezing bandwidth, low-frequency excess noise, the

approximately 2 MHz feature, frequency-dependent atomic Langevin noise, or detector-frequency response. Therefore a single gain-derived squeezing number is not substituted for $S_-(\Omega_{\text{SA}})$.

Table 15: Discrepancies relative to the documented experimental point.

Observable	Reduced model	Experiment or benchmark	Discrepancy
G_s	5.614	approximately 15.5	−63.8%
G_c	4.668	apparent power ratio 13.6	−65.7%; definitions are not fully equivalent
$\delta^*/(2\pi)$	−275 MHz in the archived reduced scan	−8 MHz measured operating point	−267 MHz; different objective and incomplete geometry make this nonpredictive
$\Delta^*/(2\pi)$	−1.50 GHz in the archived reduced scan	+0.90 GHz measured operating point	−2.40 GHz
Maximum squeezing	no physical prediction	−7.8 dB	Microscopic diffusion and detector response missing
Squeezing bandwidth	no physical prediction	approximately 3.5 MHz	Static model has no bandwidth

The point $(\Delta/2\pi, \delta/2\pi, T) = (-1.50 \text{ GHz}, -280 \text{ MHz}, 110^\circ\text{C})$ is henceforth classified only as a *numerical optimum under the archived reduced model*. It is not an experimentally established optimum.

27 Required limiting cases

Table 16: Analytic limiting-case checks for the layered model.

Limit	Required result	Status
No atom-field coupling	$\chi_{ij} \rightarrow 0$ implies $G_s \rightarrow 1$, $G_c \rightarrow 0$, $D \rightarrow 0$, and $S_- \rightarrow 1$	PASS analytically
Ideal lossless parametric amplifier	$M_q J + J M_q^\dagger = 0$, $B = D = 0$, and $G_c = G_s - 1$	PASS analytically
Bright seeded ideal amplifier	$S_- = 1/(2G_s - 1)$ in the stated balanced normalization	PASS analytically
Symmetric external efficiency	$S_-^{\text{det}} = 1 - \eta + \eta S_-^{\text{out}}$	PASS analytically
Zero crossing angle	$\theta \rightarrow 0$ removes $k\theta\sigma_v$ from the two-photon Doppler width	PASS analytically
Zero dissipation	The Langevin contribution vanishes and $TJT^\dagger = J$	PASS as a formal limit

28 Conclusions and validation status

The corrected result is deliberately layered.

- **Derived analytically:** detuning and frequency conventions, the base Faddeeva average, the constant- M_{cl} matrix exponential, the classical gain-gap identity, the local and integrated commutator identities, external-loss mixing, and the limiting cases in Table 16.
- **Formulated as finite-dimensional matrix problems:** the self-consistent pump-only Liouvillian null space and frequency-dependent weak-field resolvent. They are specified here but are not yet implemented in the production FWM path.
- **Solved numerically under the inherited pump approximation:** the Floquet block system, classical propagation, convergence audits, and the illustrative minimum-vacuum channel dilation.
- **Inherited from the previous code:** the four-hyperfine-level truncation, effective line-strength factor, several relaxation rates, archived scan ranges, and empirical coefficients listed in Appendix B.
- **Inherited or phenomenological:** $\ell_s = 0.74$, $\kappa = 0.1$, the adopted γ_{gg} , and any pump-scattering spectrum. No historical fitting record was found, so none is relabeled as a measured quantity.
- **Compared with experiment:** the corrected reduced-model gains were evaluated at the documented 600 mW, 121 °C, 12.5 mm, 0.32° point. The disagreement in Table 15 precludes a claim of quantitative gain validation.
- **Unvalidated:** the microscopic diffusion matrix, $S_-(\Omega_{SA})$, absolute spontaneous-emission noise, Raman linewidth, polarization and magnetic-field response, and the predicted optimum after a full non-collinear velocity average.

Thus the retained calculation is a semianalytic reconstruction of the *semiclassical mean-field gain model*. The old values $\xi_{finite} = -8.10$ dB and $\xi_{ideal} = -15.62$ dB are historical, gain-derived numbers and are not physically validated squeezing predictions. The report title should include intensity-difference squeezing only after $B(\Omega_{SA})$, $D(\Omega_{SA})$, detector response, and the experimental spectrum have been implemented and validated.

A Assumptions and validity domains

Table 17: Assumption and validity-domain ledger.

Layer	Assumption	Validity domain and failure mode
Atomic basis	Four hyperfine manifolds with Zeeman structure averaged into effective strengths	Hyperfine-scale qualitative trends; fails for polarization, optical pumping, dark states, and magnetic response
Pump steady state	Undepleted prescribed pump and finite-dimensional Markovian Liouvillian	Controlled only under the stated truncation and phenomenological decay channels
Optional rate model	Populations solved without coherence feedback	Weak saturation or rapid dephasing; not an exact strong-pump steady state
Weak fields	Seed and conjugate enter to first order	Below pump depletion, gain saturation, and nonlinear probe back-action

Layer	Assumption	Validity domain and failure mode
Floquet expansion	Finite N_F	Must be increased until observables converge; $N_F = 1$ fails at the archived point
Velocity average	Maxwell distribution, independent Cartesian components	Quantitative geometry requires at least the mismatch projection; the old one-dimensional average is incomplete
Classical propagation	z -independent coefficients and fixed density, temperature, and pump	Closed-form exponential; fails with longitudinal pump attenuation, density gradients, or depletion
Phase convention	Diagonal complex susceptibility plus vacuum/geometric mismatch only	Avoids double counting of $\text{Re } \chi$
Quantum propagation	Linear Gaussian Langevin channel	Requires a microscopic reservoir matrix and appropriate Markov approximation
Static dilation	Minimum-vacuum factorization of the corrected drift	Commutator diagnostic only; not an atomic-noise prediction
Detection	External path and detector losses act after the cell	Valid only for losses physically distinct from distributed atomic absorption
Electronics	Linear transfer functions and calibrated shot-noise reference	Requires measured $H_s, H_c, w, S_{\text{el}}$ at every analysis frequency

B Parameter provenance and sensitivity

The classifications below distinguish quantities measured for the cited apparatus, assumed in the reduced model, fitted or inherited phenomenologically, and algebraically derived. A dash means that the parameter does not enter that model layer.

Table 18: Measured, assumed, fitted, inherited, and derived parameters.

Parameter	Physical definition	Class and source	Procedure or allowed range	Sensitivity of G_s, G_c, S_-
$L = 12.5 \text{ mm}$	Vapor interaction length	Measured; experimental cell [1]	Use manufacturing/calibration uncertainty when available	Gains depend through $\exp(M_{\text{cl}}L)$; spectrum also depends on integrated diffusion
$T = 121 \text{ }^\circ\text{C}$	Cell-reservoir operating temperature	Experimentally measured operating point [1]	Not a fitted model optimum	Strong density and Doppler dependence; must be rescanned

Parameter	Physical definition	Class and source	Procedure or allowed range	Sensitivity of G_s, G_c, S_-
$P_p = 600 \text{ mW}$, $P_s = 8 \mu\text{W}$	Incident pump and seed powers used for squeezing	Experimentally measured operating point [1]	Seed was approximately $10 \mu\text{W}$ in the apparatus description; $8 \mu\text{W}$ for the cited squeezing trace	Pump changes the steady state; seed sets bright-beam linearization and SNL
$w_p = 530 \mu\text{m}$, $w_s = 330 \mu\text{m}$ $\theta = 0.32^\circ$	$1/e^2$ intensity radii, not diameters Pump–seed crossing angle	Measured [1] Measured geometry [1]	Use the reported beam convention Zero-angle check required	Set Rabi frequency and spatial overlap Produces 1.36 MHz rms angular Raman broadening
$\Delta/(2\pi) = +0.90 \text{ GHz}$, $\delta/(2\pi) = -8 \text{ MHz}$	Experimental optical and two-photon operating detunings	Experimentally measured operating point [1]	Not identified with the archived numerical optimum	Model evaluation gives $G_s = 5.614$, $G_c = 4.668$
$\Gamma/(2\pi) = 5.75 \text{ MHz}$	Natural optical linewidth used in the model	Literature benchmark	Ordinary-frequency value converted once to angular rate	Sets optical resolvents and Voigt width
$\sigma_v = 193.7 \text{ m/s}$ at 110°C ; 196.5 m/s at 121°C	One-axis thermal rms speed at the archived and literature temperatures	Derived from Maxwell statistics	$\sigma_v = \sqrt{k_B T/m}$	Sets both one-photon and mismatch-projected Doppler widths (1.36 and 1.38 MHz, respectively)
$\ell_s = 0.74$	Effective line-strength or overlap multiplier in the archived response	Inherited from code; phenomenological coefficient with unknown fit provenance	Original fitting record unavailable; audit range 0.666–0.814	$\partial \ln G_s / \partial \ln \ell_s \simeq 3.53$, $\partial \ln G_c / \partial \ln \ell_s \simeq 4.29$; physical S_- unavailable
$\kappa = 0.1$	Coupling of absorbed pump fraction to detected pump-scattering noise	Inherited from code; assumed phenomenological coefficient	No documented fit; illustrative range 0–0.2 pending apparatus calibration	No effect on G_s, G_c ; ansatz noise is linear in κ , but normalized $S_-(\Omega)$ requires F_{ps} and detector calibration
$\gamma_{gg}/(2\pi) = 100 \text{ kHz}$	Effective ground-state decoherence rate	Inherited from code; assumed phenomenological rate	Audit range 90–110 kHz; independent linewidth fit absent	Across this range $G_s = 5.6138 \rightarrow 5.6033$ and $G_c = 4.6631 \rightarrow 4.6573$; physical S_- unavailable
$1 - \eta_{\text{path}} = 5.5(2)\%$	Post-cell optical-path loss	Measured [1]	Allowed range 5.3–5.7%; not cell absorption	No effect on gains; for symmetric loss $\partial S_{\text{det}} / \partial \eta = S_{\text{out}} - 1$
$1 - \eta_{\text{det}} = 8.0(2)\%$	Detector and remaining system inefficiency	Measured [1]	Allowed range 7.8–8.2%	No effect on gains; combines multiplicatively with path efficiency
$\eta_{\text{ext}} = 0.8694$	Illustrative total external efficiency 0.945×0.920	Derived from measured losses	The paper also quotes total loss 13.5(2)%; definitions must be reconciled	Maps S_{out} to $1 - \eta_{\text{ext}} + \eta_{\text{ext}} S_{\text{out}}$
N_F	Retained Floquet sideband order	Numerical truncation	$N_F = 2$ converged at the tested point; use $N_F \geq 3$ for reported scans	$N_F = 1$ changes gains by 13–17% and reverses the gain-gap sign

Parameter	Physical definition	Class and source	Procedure or allowed range	Sensitivity of G_s, G_c, S_-
$\ell_{\text{eff}} = 0.74(7/144)$	Archived combined line/population factor	Derived from inherited factors, with normalization unresolved	$\ell_{\text{eff}} = 0.0359722$; see Appendix C	Must not be applied if the same population and Zeeman weights are already in ρ_{ss}

No entry involving ℓ_s , κ , or the adopted γ_{gg} is described as exact. Their source records and apparatus-level fitting likelihoods remain required metadata.

C Dipole and population normalization ledger

Let $d_{D1} = \langle J_e || er || J_g \rangle$ be the fine-structure reduced matrix element. The archived coefficient is not the polarization-summed hyperfine strength. It is the following *AutoOD* $q = 0$ convention:

$$\mathcal{S}_0(F_g, F_e) = \sum_{m=-\min(F_g, F_e)}^{+\min(F_g, F_e)} \begin{pmatrix} F_e & 1 & F_g \\ m & 0 & -m \end{pmatrix}^2, \quad (39)$$

$$C_F^2 = (2F_g + 1)(2F_e + 1)(2J_g + 1)(2J_e + 1)(2L_g + 1)\mathcal{S}_0(F_g, F_e) \\ \times \left\{ \begin{matrix} J_g & J_e & 1 \\ F_e & F_g & I \end{matrix} \right\}^2 \left\{ \begin{matrix} L_g & L_e & 1 \\ J_e & J_g & S \end{matrix} \right\}^2. \quad (40)$$

For $I = 5/2$, $S = 1/2$, $L_g = 0$, $L_e = 1$, and $J_g = J_e = 1/2$, $\mathcal{S}_0 = 1/3$ and this gives the stored exact values

$$C_F^2 = \{10, 35, 35, 28\}/81$$

for $(F_g, F_e) = (2, 2), (2, 3), (3, 2), (3, 3)$, respectively.

The AutoOD orbital reduced dipole and the fine-structure reduced dipole used by GABES obey $|d_{\text{orb}}|^2 = 3|d_{D1}|^2$ in this D1 convention. Therefore

$$|d_{\text{eff}}|^2 = C_F^2 |d_{\text{orb}}|^2 = 3C_F^2 |d_{D1}|^2,$$

and the dimensionless amplitude multiplier that accompanies d_{D1} is

$$a_{ge} = \sqrt{3C_F^2}. \quad (41)$$

This derivation explains the code factor but also its limitation: Eq. (40) is a $q = 0$, Zeeman-averaged absorption convention, not a universal incoherent sum over $q = 0, \pm 1$. A polarization-resolved Liouvillian must replace it with explicit Clebsch–Gordan amplitudes and must not multiply by a_{ge} again.

The archived effective factor was assembled as

$$\ell_{\text{eff}} = \ell_s \frac{p_{F=3}}{2(2I + 1)} = 0.74 \times \frac{7}{144} = 0.0359722. \quad (42)$$

Here ℓ_s is the phenomenological line/overlap multiplier, $p_{F=3} = 7/12$ is the assumed hyperfine statistical population, and $2(2I + 1) = 12$ for $I = 5/2$ is the adopted manifold normalization. This decomposition is documented, but not independently derived from the present Liouvillian.

The normalization rule for future implementations is:

1. use the fine-structure reduced dipole matrix element once;
2. use either explicit Clebsch–Gordan matrix elements or the averaged a_{ge} , never both;
3. include isotope abundance once in N ;

4. obtain ground-manifold population either from ρ_{ss} or an external p_F , never both;
5. include spontaneous-emission branching ratios in the collapse operators, not again in susceptibility amplitudes; and
6. apply external optical efficiencies only in the detection layer.

Table 19: Code-path normalization audit for the current reduced FWM solver.

Factor	Where it enters	Verdict
Reduced D1 dipole d_{D1}	Once in the macroscopic conversion in <code>gabes/observables.py</code>	no duplicate found
Hyperfine/Zee-man strength	$\sqrt{3C_F^2}$ in the weak-field drive and once in polarization readout; their product gives $3C_F^2$. It is not also present in <code>physical_coupling_norm</code> .	no algebraic duplicate found
Isotope abundance	The FWM path uses the pure- ⁸⁵ Rb density from <code>hyperfine.number_density</code> ; it does not multiply the natural 0.7217 abundance.	no duplicate in this path
Ground-manifold population	ρ_{ss} is trace normalized over the two lumped ground manifolds, while <code>physical_coupling_norm</code> also supplies $p_F/[2(2I + 1)]$.	potential duplicate
Spontaneous branching	The C_F^2 -normalized decay fractions occur in Liouvillian collapse channels; no additional branching multiplier is applied in <code>chi_phys</code> .	no duplicate found
Residual ℓ_s	Multiplies the macroscopic coupling once.	inherited, not derived

The ground-population row prevents a blanket verification claim. Because the archived code combines ℓ_s and 7/144 without a complete representative- state derivation, repeated ground-state population weighting remains unresolved even though the other listed factors have single, identifiable code paths. Equation (42) must not be carried into a trace-normalized explicit-state Liouvillian until that row is reconciled.

D Four-level model limitations and unresolved physics

The four-hyperfine-level model is suitable for qualitative gain structure, reduced hyperfine-scale intuition, and comparison with the existing reduced simulation. It is not automatically sufficient for absolute squeezing, polarization-sensitive effects, magnetic-field response, quantitative spontaneous-emission noise, or accurate Raman linewidths.

Physics outside the current basis and reservoir model includes:

- polarization-dependent Clebsch–Gordan pathways and explicit $\sigma^\pm$ and π selection rules;
- Zeeman optical pumping, ground-state orientation and alignment, dark-state formation, and magnetic-field-dependent coherences;
- interference among distinct m_F pathways and polarization-dependent spontaneous emission;
- spatial pump depletion, longitudinal attenuation, transverse Gaussian averaging, diffraction, mode mismatch, and cell-window etalons;
- buffer-gas or wall-collision dynamics, transit-time distributions, radiation trapping, and non-Markovian reservoirs;
- multidimensional velocity-changing collisions and the full non-collinear Raman Doppler distribution;
- a microscopic frequency-dependent atomic diffusion matrix, excess fluorescence, and atom–reservoir correlations;

- detector transfer functions, electronic cross-talk, common-mode rejection, and the apparatus-specific 2 MHz laser-electronics feature; and
- an independently measured shot-noise calibration and gain definition matched exactly to the theoretical modes.

These are not small corrections by default. In particular, the observed gain discrepancy and the 1.36 MHz angular Doppler scale show that omitted geometry and level structure can be quantitatively important.

E Change log mapped to the original report

Line numbers in Table 20 refer to the pre-revision version of this file. The equation labels are retained here only as audit anchors; the revised equations have new labels.

Table 20: Correction-by-correction change log.

No.	Original location	Revision	Validation or remaining status
1	Noise section, Eqs. (old <code>io</code> , <code>S</code>), lines 597–628	Removed the gain-only Bogoliubov interpretation and old cov_0 formula; introduced Eqs. (28)–(33) with reservoir channels	Commutator identities pass for the mathematical dilation; microscopic $D(\Omega)$ remains pending
2	Beat-frequency subsection and scalar noise result, lines 207–214 and 597–664	Separated Ω_{beat} from Ω_{SA} and made M_q, B, D, V, S_- frequency dependent	Static model limitations stated; spectrum prediction pending
3	Carrier rate equations and Sylvester block, old Eqs. <code>rateeq-sylv</code> , lines 255–305	Replaced the claimed strong-pump solution by a trace-constrained Liouvillian null-space problem; retained rate equations only as an explicitly controlled approximation	Self-consistent finite-dimensional formulation supplied; implementation requires code migration
4	Floquet expansion, old Eq. <code>floquet</code> , lines 216–247	Repeated $N_F = 1, 2, 3$ and reported gains, gap, response phases, and δ^*	$N_F = 1$ fails; $N_F = 2$ and 3 agree at the tested point
5	One-dimensional Doppler section, old lines 455–494	Added vector detunings, mismatch-projected two-dimensional average, and 1.36 MHz angular Raman width	One-dimensional quadrature converged; full geometry not yet implemented
6	Input and summary tables, old lines 113–188 and 667–742	Classified the point with $\Delta/(2\pi) = -1.50$ GHz, $\delta/(2\pi) = -280$ MHz, and $T = 110^\circ\text{C}$ as an archived-model optimum and evaluated the reduced model at the documented experimental point	Gain and optimum discrepancies reported separately

No.	Original location	Revision	Validation or remaining status
7	Abstract, summary, and error-budget claims, old lines 55–67 and 667–718	Recast agreement as algebraic and numerical implementation consistency and introduced a five-level validation hierarchy	No claim of zero physical error or experimental validation remains
8a	Macroscopic coupling discussion, old lines 174–186	Corrected $[\bar{\chi}] = s$ and $[Nd^2/(\epsilon_0\hbar)] = s^{-1}$	Dimensional audit in Table 5
8b	Voigt identity, old Eq. voigt , lines 472–486	Removed the erroneous $1/k$ prefactor: $\text{Im}\langle(x_0 - kv + i\Gamma)^{-1}\rangle = -\pi V$	Dimensions rechecked
8c	Phase-matching paragraph, old lines 520–524	Replaced $f_c = 2\Delta - f_{\text{seed}}$ by $\omega_c = 2\omega_p - \omega_s$ with explicit atomic references	Frequency audit supplied
8d	Equations and captions throughout	All detunings and decay rates are angular quantities; displayed values use $x/(2\pi)$ in ordinary-frequency units	Frequency audit supplied
9	Old Eqs. Mentries and phasematch , lines 498–524	Selected Option A: complex diagonal susceptibilities plus vacuum/geometric Δk_z only	Derivation in Part IV; no $\text{Re } \chi$ appears twice
10	Arm-OD and detector-noise sections, old lines 614–645	Kept atomic loss inside M_q, B, D and applied only physically external efficiency after the cell	Loss ledger in Table 3
11	Old lines 167, 180–186, 627–645, and parameter summary	Classified $\ell_s, \kappa, \gamma_{gg}$, and external losses; documented definitions, sources, ranges, fitting status, and sensitivities	Missing historical fit records are marked unresolved
12	Four-hyperfine-level model and old CG normalization, lines 88–106	Added limitations and normalization ledgers, including $a_{ge} = \sqrt{3C_F^2}$ and $\ell_{\text{eff}} = 0.74(7/144)$	Potential repeated ground-population weighting explicitly unresolved
13	Mean-field, Faddeeva, propagation, and gain-gap sections	Retained the valid classical response chain, corrected base Faddeeva integral, constant-matrix exponential, classical gain identity, beat convention, and beam radii	Each retained result now carries its assumptions
14	Whole-report organization	Reorganized the text into Definitions, Pump steady state, Weak response, Semiclassical propagation, Quantum propagation, Detection, and Validation	Complete
15	Previously absent	Added six limiting-case checks	Analytic checks pass; microscopic numerical limits await implementation

No.	Original location	Revision	Validation or remaining status
16	Original title and conclusion, old lines 55–75 and 667–726	Adopted a provisional mean-field title and invalidated the physical status of -8.10 dB and -15.62 dB	Conclusion now separates derived, solved, inherited, fitted, compared, and unvalidated content
17	Previously absent as an auditable set	Added convergence, commutator, comparison, provenance, dimensional, frequency, assumptions, limitations, and change-log deliverables	Machine-readable convergence artifacts accompany the report

F Reproducibility and acceptance audit

The Floquet, velocity-grid, corrected Option-A, and commutator-diagnostic tables in this report are generated by `analysis/analytic_reconstruction/convergence_audit.py`. Its human-readable and machine-readable outputs are `generated/convergence_audit.md` and `generated/convergence_audit.json`. The script records the operating points, truncations, velocity grids, corrected phase convention, channel dilation, and residual norms used above.

Table 21: Acceptance-criterion status of this provisional revision.

Criterion	Status
Canonical output commutators with the supplied reservoir channels	PASS conditionally for the $Q \propto I$ diagnostic; calibrated canonical scaling pending
Frequency-resolved observable $S_-(\Omega_{SA})$	formulated; microscopic result not yet available
Atomic dissipation accompanied by Langevin noise	required by construction; microscopic B, D not yet available
No repeated internal absorption in the detection layer	PASS
No repeated dispersive phase in Δk_z	PASS
Self-consistent pump steady-state formulation	PASS; code migration pending
Converged Floquet truncation	PASS at the tested point for $N_F \geq 2$
Non-collinear two-photon Doppler broadening	quantified and formulated; numerical 2D implementation not yet available
Consistent angular-frequency convention	PASS
Correct susceptibility and Voigt dimensions	PASS
Shared-model agreement not called experimental validation	PASS
Empirical-parameter classification	PASS
Ground-population normalization counted only once	not yet available ; potential duplicate isolated in Table 19
Mean-field and quantum layers separated	PASS

Accordingly, this document is acceptable as a corrected *mean-field reconstruction and quantum-model specification*. It does not yet satisfy the stronger acceptance condition for a physically validated

squeezing calculation. That condition requires the pending microscopic $B(\Omega_{\text{SA}})$, $D(\Omega_{\text{SA}})$, two-dimensional velocity integration, and detector transfer functions.

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